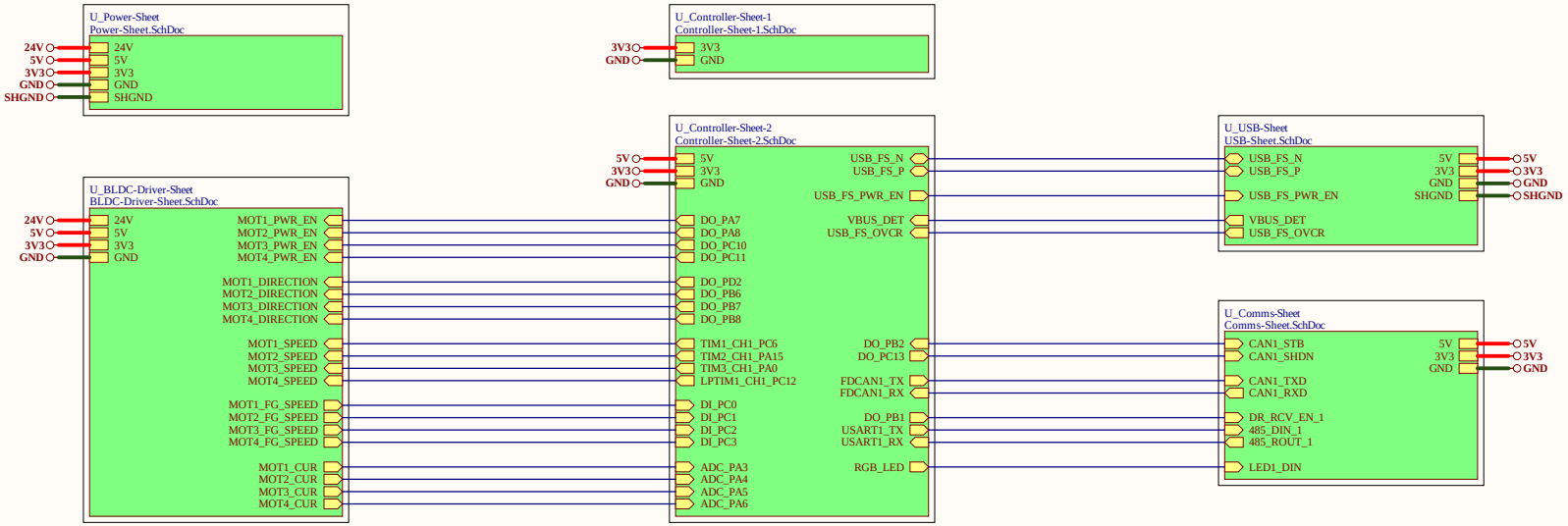
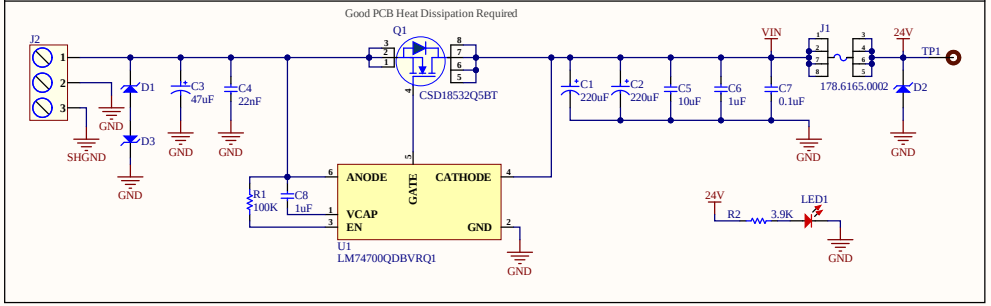
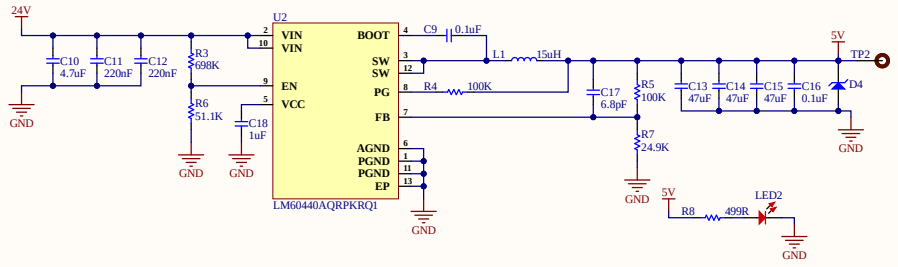


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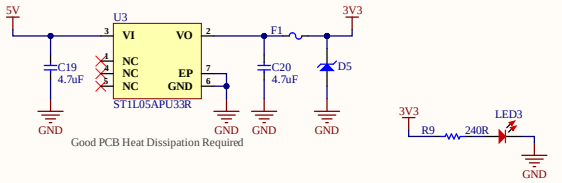
24V INPUT

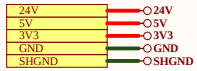


5V BUCK CONVERTER



3.3V LINEAR REGULATOR

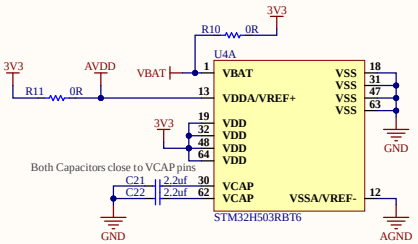




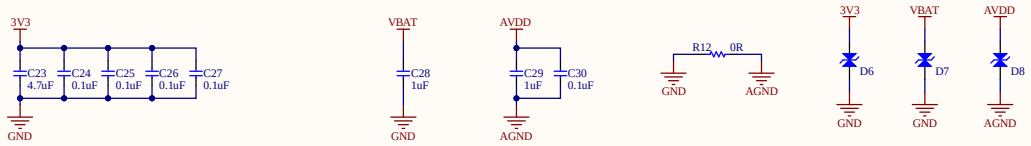
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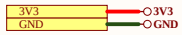
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CONTROLLER POWER SHEET



MCU DECOUPLING CAPACITORS

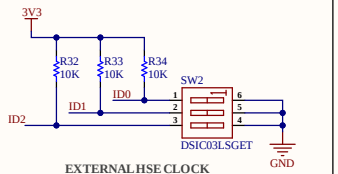




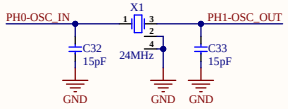
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File: BLDC-Driver-v2.0.PrjPcb	
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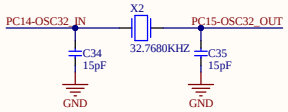
CONTROLLER IO SHEET

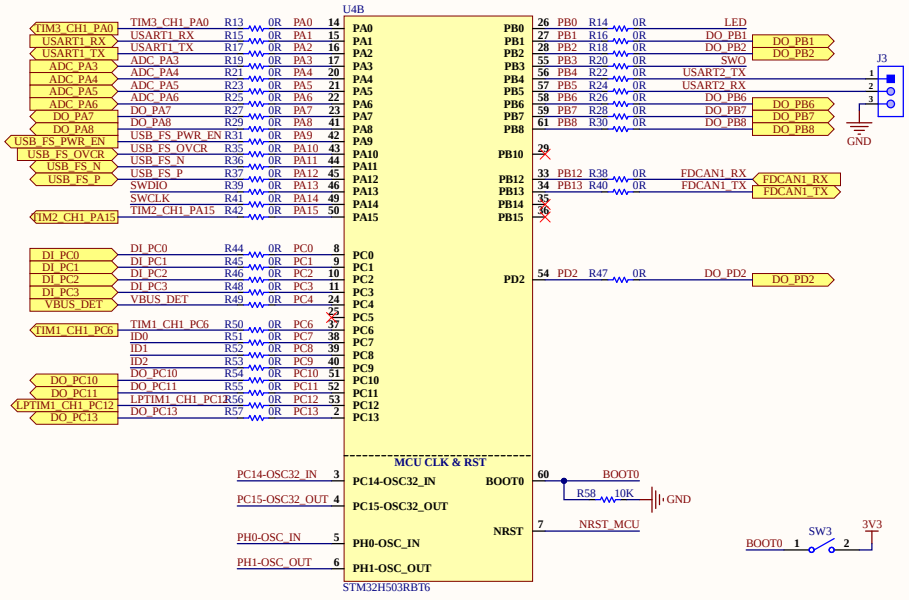


EXTERNAL HSE CLOCK

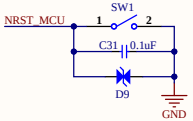


EXTERNAL LSE CLOCK

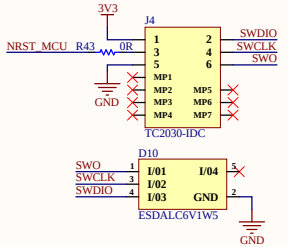




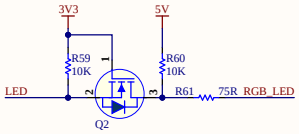
RESET FUNCTION



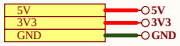
PROGRAMMING TAG CONNECT



RGB LED LEVEL TRANSLATOR



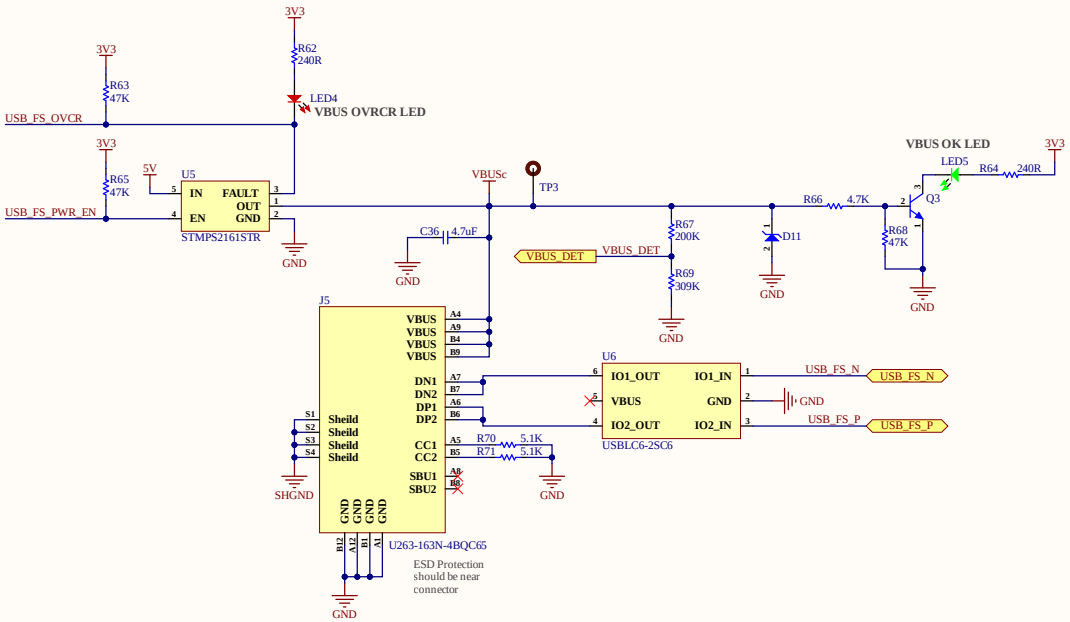




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Title:	Controller-Sheet-2.SchDoc
File:	BLDC-Driver-v2.0.PrjPcb
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USB

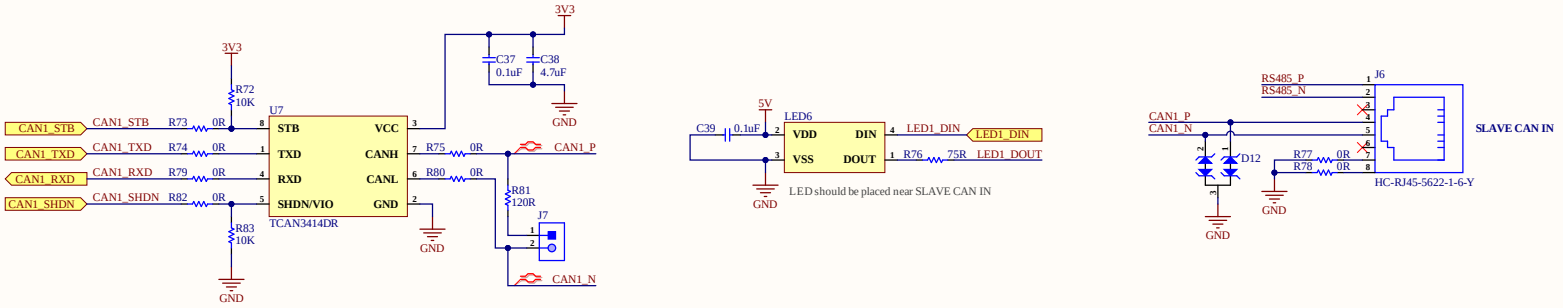


USB_FS_PWR_EN
USB_FS_OVCR

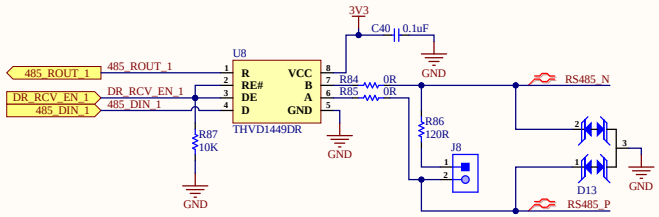
5V
3V3
GND
SHGND

Title:	USB-Sheet.SchDoc
File:	BLDC-Driver-v2.0.PrjPcb
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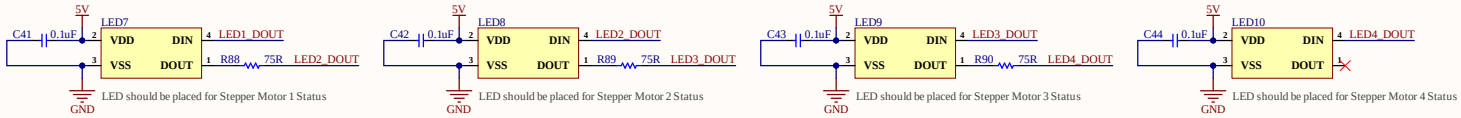
CAN

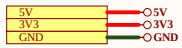


RS485



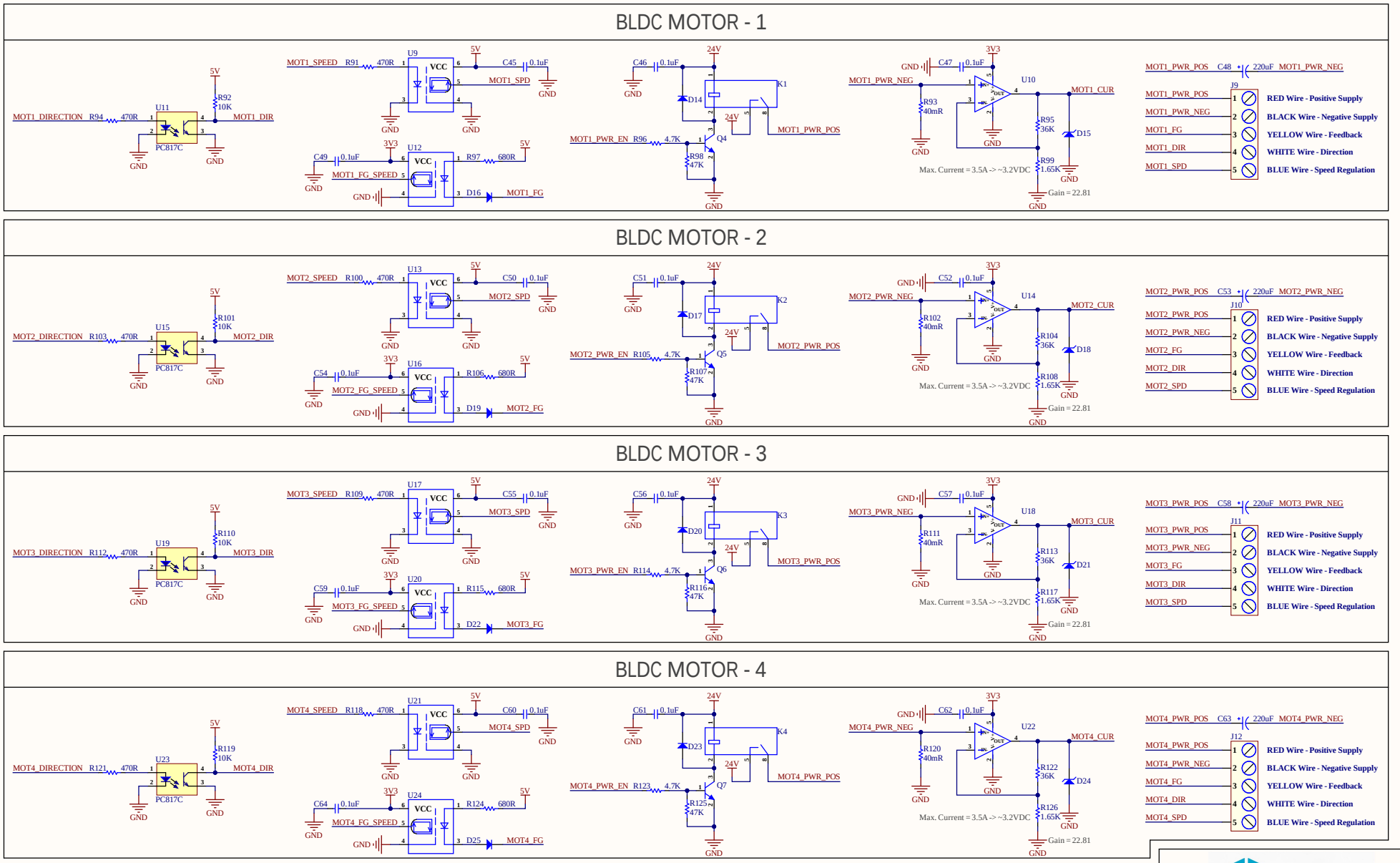
LEDs





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Title: Comms-Sheet.SchDoc	
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BLDC MOTOR - 1
MOT1_DIRECTION R94 470R U11 PC817C R92 10K MOT1_DIR
MOT1_SPEED R91 470R U9 VCC C45 0.1uF MOT1_SPD
C49 0.1uF U12 VCC R97 680R MOT1_FG_SPEED D16 MOT1_FG
C46 0.1uF D14 K1 MOT1_PWR_EN R96 4.7K Q4 R98 47K MOT1_PWR_POS
MOT1_PWR_NEG C47 0.1uF U10 R93 40mR R95 36K R99 1.65K D15 MOT1_CUR
Max. Current = 3.5A -> ~3.2VDC
Gain = 22.81
MOT1_PWR_POS C48 220uF MOT1_PWR_NEG
J9
MOT1_PWR_POS RED Wire - Positive Supply
MOT1_PWR_NEG BLACK Wire - Negative Supply
MOT1_FG YELLOW Wire - Feedback
MOT1_DIR WHITE Wire - Direction
MOT1_SPD BLUE Wire - Speed Regulation
BLDC MOTOR - 2
MOT2_DIRECTION R103 470R U15 PC817C R101 10K MOT2_DIR
MOT2_SPEED R100 470R U13 VCC C50 0.1uF MOT2_SPD
C54 0.1uF U16 VCC R106 680R MOT2_FG_SPEED D19 MOT2_FG
C51 0.1uF D17 K2 MOT2_PWR_EN R105 4.7K Q5 R107 47K MOT2_PWR_POS
MOT2_PWR_NEG C52 0.1uF U14 R102 40mR R104 36K R108 1.65K D18 MOT2_CUR
Max. Current = 3.5A -> ~3.2VDC
Gain = 22.81
MOT2_PWR_POS C53 220uF MOT2_PWR_NEG
J10
MOT2_PWR_POS RED Wire - Positive Supply
MOT2_PWR_NEG BLACK Wire - Negative Supply
MOT2_FG YELLOW Wire - Feedback
MOT2_DIR WHITE Wire - Direction
MOT2_SPD BLUE Wire - Speed Regulation
BLDC MOTOR - 3
MOT3_DIRECTION R112 470R U19 PC817C R110 10K MOT3_DIR
MOT3_SPEED R109 470R U17 VCC C55 0.1uF MOT3_SPD
C59 0.1uF U20 VCC R115 680R MOT3_FG_SPEED D22 MOT3_FG
C56 0.1uF D20 K3 MOT3_PWR_EN R114 4.7K Q6 R116 47K MOT3_PWR_POS
MOT3_PWR_NEG C57 0.1uF U18 R111 40mR R113 36K R117 1.65K D21 MOT3_CUR
Max. Current = 3.5A -> ~3.2VDC
Gain = 22.81
MOT3_PWR_POS C58 220uF MOT3_PWR_NEG
J11
MOT3_PWR_POS RED Wire - Positive Supply
MOT3_PWR_NEG BLACK Wire - Negative Supply
MOT3_FG YELLOW Wire - Feedback
MOT3_DIR WHITE Wire - Direction
MOT3_SPD BLUE Wire - Speed Regulation
BLDC MOTOR - 4
MOT4_DIRECTION R121 470R U23 PC817C R119 10K MOT4_DIR
MOT4_SPEED R118 470R U21 VCC C60 0.1uF MOT4_SPD
C64 0.1uF U24 VCC R124 680R MOT4_FG_SPEED D25 MOT4_FG
C61 0.1uF D23 K4 MOT4_PWR_EN R123 4.7K Q7 R125 47K MOT4_PWR_POS
MOT4_PWR_NEG C62 0.1uF U22 R120 40mR R122 36K R126 1.65K D24 MOT4_CUR
Max. Current = 3.5A -> ~3.2VDC
Gain = 22.81
MOT4_PWR_POS C63 220uF MOT4_PWR_NEG
J12
MOT4_PWR_POS RED Wire - Positive Supply
MOT4_PWR_NEG BLACK Wire - Negative Supply
MOT4_FG YELLOW Wire - Feedback
MOT4_DIR WHITE Wire - Direction
MOT4_SPD BLUE Wire - Speed Regulation
MOT1_DIRECTION MOT2_DIRECTION MOT3_DIRECTION MOT4_DIRECTION
MOT1_SPEED MOT2_SPEED MOT3_SPEED MOT4_SPEED
MOT1_FG_SPEED MOT2_FG_SPEED MOT3_FG_SPEED MOT4_FG_SPEED
MOT1_PWR_EN MOT2_PWR_EN MOT3_PWR_EN MOT4_PWR_EN
MOT1_CUR MOT2_CUR MOT3_CUR MOT4_CUR
24V 5V 3V3 GND
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